Experiment 10: Transformation-Based Part-of-Speech Tagging in Python

1. Aim

To write a Python program to implement transformation-based tagging using a set of transformation rules.

2. Procedure

1. Define a tokenized sentence and its initial baseline POS tags.
2. Define a contextual transformation rule: change `'NN'` to `'VB'` if the previous tag is `'TO'`.
3. Iterate through the tagged words and check if the preceding tag satisfies the rule.
4. Apply the tag transformation where the rule condition is met.
5. Display the initial tags and the final transformed POS tags.

3. Program

```
words = ["I", "want", "to", "race"]
tags = ["PRP", "VBP", "TO", "NN"]

print("Initial Tags:", list(zip(words, tags)))

for i in range(1, len(tags)):
    if tags[i] == "NN" and tags[i-1] == "TO":
        tags[i] = "VB"

print("Transformed Tags:", list(zip(words, tags)))
```

4. Output

```
Initial Tags: [('I', 'PRP'), ('want', 'VBP'), ('to', 'TO'), ('race',
'NN')]
Transformed Tags: [('I', 'PRP'), ('want', 'VBP'), ('to', 'TO'), ('race',
'VB')]
```

5. Result

Thus, the Python program was successfully executed to implement transformation-based tagging by applying a transformation rule.